

SMART KOPARGOAN HACKATHON

TITLE PAGE



- **Problem Statement ID – skh021**
- **Problem Statement Title- GPS-Based Smart Garbage Collection and Monitoring System**
- **Theme- Open Innovation**
- **PS Category- Software**
- **Team ID- JF9NXM**
- **Team Name- Navotkarsh**

CleanRoute -Smart Waste Management Platform

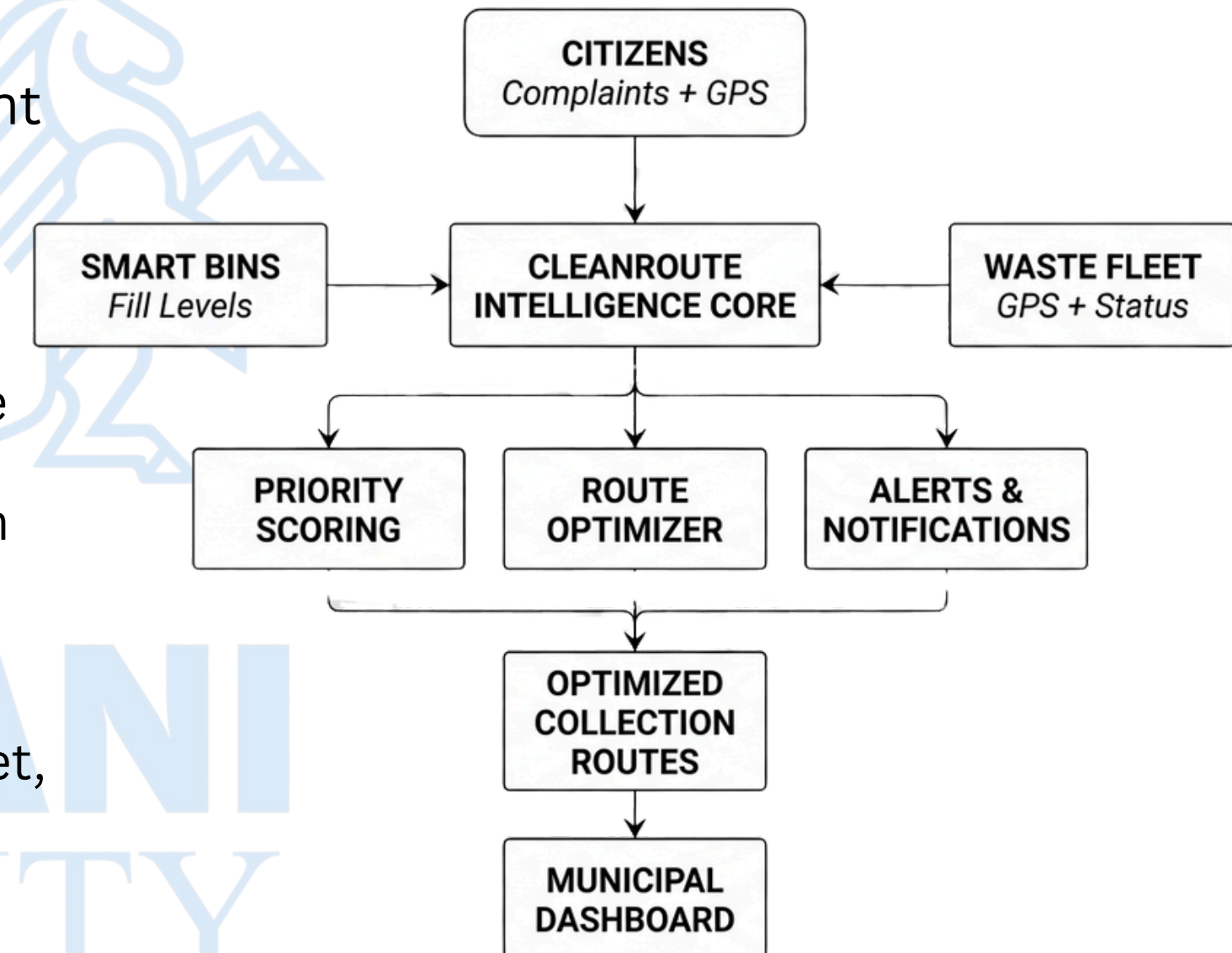
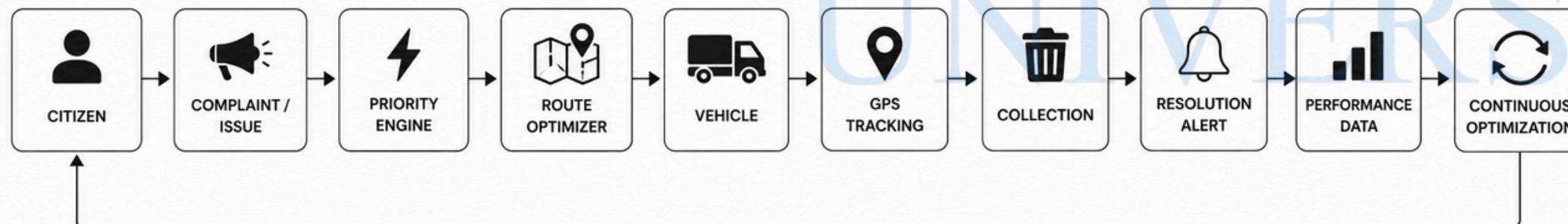


Idea/Solution:

CleanRoute is an integrated smart waste-management platform that **connects** citizens, municipal authorities, waste vehicles, and collection points on a single system. It **combines** real-time GPS fleet tracking, intelligent route optimization, bin monitoring, priority-based complaint management, and alerts to enable faster, more efficient, and accountable waste collection.

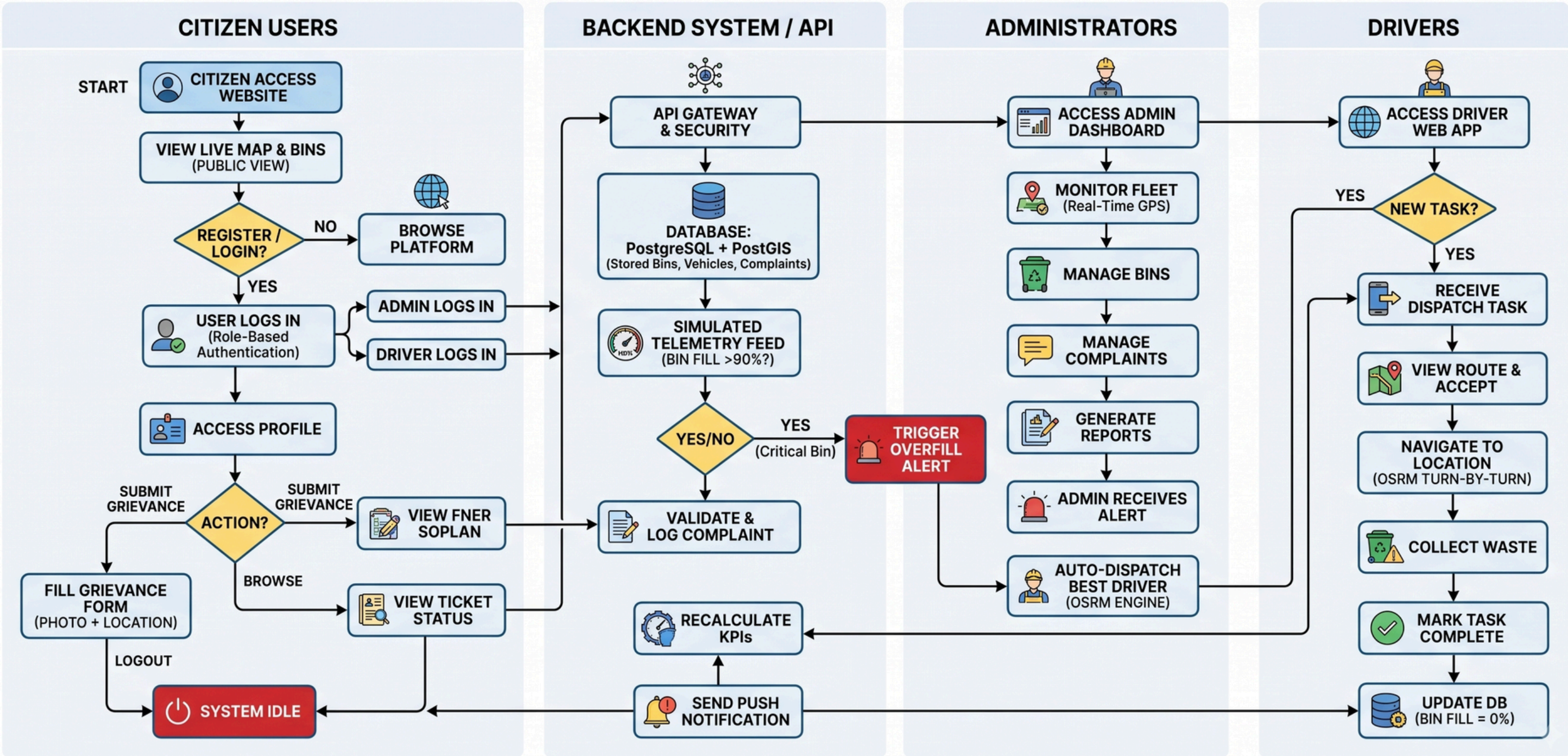
Key Components :

- GPS Vehicle Tracking: Real-time visibility of garbage-truck locations, route progress and fleet activity.
- AI-Powered Route Optimization: Generates efficient daily routes based on collection points, vehicle capacity and operational constraints.
- Bin + Overflow Monitoring: Tracks fill levels and automatically prioritizes high-fill bins for collection.
- End-to-End Complaint Management: Citizens report issues, receive a ticket, and track resolution while administrators assign and monitor actions.





SMART WASTE MANAGEMENT PLATFORM: INTEGRATED WORKFLOW



Layer / Component	Technology
Backend Framework	Node.js / NestJS
Geospatial Database	PostgreSQL + PostGIS
In-Memory Cache	Redis
IoT Telemetry	MQTT (EMQX)
Data Streaming	Apache Kafka
Route Optimization	Google OR-Tools
Task Queue	BullMQ / Celery
Real-Time Engine	Socket.io
Web Frontend	React.js
Map Rendering	Mapbox GL JS
State Management	Zustand
Push Notifications	Firebase Cloud Messaging
Cloud Hosting	AWS (EC2, RDS)



FEASIBILITY AND VIABILITY

From Prototype To City-Wide Deployment

● TECHNICALY FEASIBLE	● Key CHALLENGES	● MITIGATION
GPS + Maps	Route complexity	Simulated API
Smartphones	GPS accuracy	Offline/PWA
Scalable	IoT hardware	JS fallback
Cloud	Route complexity	Redis/SSE

Scalable Roadmap

Pilot → Wards → City→ Multi-City
The platform can scale from a pilot ward to multiple wards, city-wide deployment.

Viable Because

\$0 MVP
\$50–100/month / city
SaaS + Govt + White-label

MVP FEASIBILITY

GPS Tracking
Route Optimization
Bin Monitoring

Smarter Routes • Cleaner Cities • Better Tomorrow



IMPACT AND BENEFITS

Fleet Supervisors

- Instant view of truck locations and route progress
- Automatic escalation of breakdowns and deviations
- Daily route plans auto-generated — no manual planning

Municipal Administrators

- Real-time visibility into entire fleet from one screen
- Data-driven decisions with **auto-generated reports**
- Reduced operational cost → budget savings redirected to other services

Field Workers / Drivers

- Mobile-friendly driver app with route guidance
- Auto-synced routes → no confusion about next stop
- Simplified reporting

Citizens Transparent

- complaint system with ticket tracking
- Fewer overflowing bins near homes/markets
- Push notifications when complaint is resolved
- Trust in local government increases



RESEARCH AND REFERENCES

Title	Link
1.Solid Waste Management Rules, 2016	Source
2.Swachh Bharat Mission – Urban 2.0	Source
3.Maharashtra Pollution Control Board – Solid Waste Management	Source
4.Municipal Solid Waste Management: A Review of Optimization Approaches	Source
5.Vehicle Routing Model for Urban Solid Waste Management – Bilaspur Case Study	Source
6.PostGIS – Spatial Database Extension for PostgreSQL	Source